LANDON MCNEIL

Test & Validation Engineer — Robotics & Physical AI
Newark, CA · +1-801-603-8085 · mcn.landon@gmail.com · https://lens-thinkering.github.io/

Test engineer skilled at designing fixtures, writing automations, and conducting root-cause investigations to get complex electromechanical hardware to ship. Three years validating high-voltage systems from single cell to full vehicle — now building robotics hardware and AI-driven test tooling on my own time.

Experience

Lucid Motors — Battery Test Engineer II | Newark, CA | Jul 2024 – Present

- Full-stack test engineering - Design and build custom test fixtures and data-acquisition rigs in CATIA, execute the tests, create the automation scripts using Python, and process the data. Test results drove design changes, regulatory certification, and safety validation.
- Built a PID-controlled thermal-initiation and DAQ system from scratch, making previously manual tests precise and repeatable.
- Lead root-cause investigations on complex hardware failures with mechanical, electrical, and simulation engineers, closing the loop from diagnosis to design fix.

BYU Racing Formula SAE Electric — Battery & High-Voltage Team Lead | Provo, UT | Aug 2022 – Jun 2024

- Led a 10-person team through the full design-build-test cycle of BYU Racing's first competitive 400V battery system, owning integration across battery, powertrain, and chassis. Passed electrical technical inspection on the first attempt.
- Defined and ran the thermal and cycling tests that certified the pack safe and race-ready.

Hypercraft — Battery Engineering Intern | Provo, UT | Nov 2023 – May 2024

- Took a large-truck battery pack from concept to detailed CAD in two months and designed validated fusible links that improved module fault tolerance.

Education

Brigham Young University — B.S. Mechanical Engineering, GPA 3.84 | 2018 – 2024

Skills

Test & automation: Python test automation, instrument control (SCPI, CAN bus), data analysis, root-cause analysis, DFMEA
CAD & fabrication: CATIA/3DX, SolidWorks, Siemens NX, 3D printing, manual machining
Other: C++, MATLAB, Git, LabVIEW · High-Voltage Safety Trained, NFPA 70E Certified

Projects

- **SO-101 Robotic Arm (Hugging Face LeRobot)** — Built and trained a low-cost arm for imitation learning; added strain-gauge grip-force sensing (instrumentation amp + Arduino) and ran training campaigns on the LeRobot framework. [https://lens-thinkering.github.io/#so-101]
- **Lab-Equipment MCP Server (Python)** — Built a server that gives an AI agent safe, real-time SCPI control of bench power supplies over serial and LAN, with a safety layer for automated hardware-in-the-loop testing. [https://github.com/lens-thinkering/BK-Precision-MCP]